

Inheritance

Inheritance: Transmission of genetic information from generation to generation.

Gene: Length of DNA that codes for specific protein

Allele: An alternative form of gene

Phenotype: Observable features of an organism

Genotype: The genetic makeup of an organism, the alleles that are present

Dominant allele: An allele that is expressed if it is present in the genotype.

Recessive allele: An allele that is only expressed when there is no dominant allele of the gene is present in the genotype.

Homozygous: Having two identical alleles of a particular gene

Heterozygous: Having two different alleles of a particular gene.

Co-dominance: A situation in which both alleles in a heterozygous organism contribute to the phenotype

Sex-linked characteristic: A feature which the gene responsible is located on a sex chromosome, and this makes the characteristic more common in one sex than the other.

Stem cells: Unspecialized cells that divide by mitosis to produce daughter cells that can become specialized for specific functions

Color blindness: A sex-linked disease.

	Phenotype	Genotype
Male	Normal vision	$X^B y$
Male	Color-blind	$x^b y$
Female	Normal vision	$X^B X^B$
Female	Color blind	$x^b x^b$
Female	Carrier	$X^B x^b$

Sex-linked diseases characteristics:

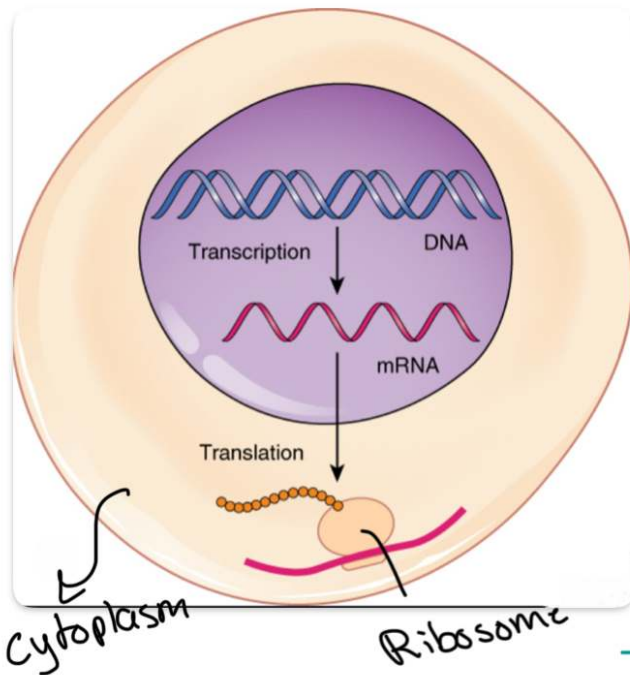
- More frequent in males than females
- Allele which causes disease is located on sex-chromosome
- The son inherits the disease from his mother
- Only females can be carriers

Protein Synthesis

*DNA controls the cell function by controlling the production of proteins: Enzymes, membrane carriers, and receptors in neurotransmitters.

Sequence of bases in genes determines the sequence of amino acids used to make a specific protein.

Sequence of bases on DNA $\rightarrow$ Sequence of amino acids $\rightarrow$ Protein



- The gene coding for the protein remains in the nucleus as DNA is a large molecule.
- mRNA is a copy of the gene
- mRNA molecules are made in the nucleus and move to the cytoplasm
- mRNA passes through ribosomes, and attaches itself to it.
- The ribosome assembles amino acids into protein molecules

Test cross: Should be with homozygous recessive:

If you know the phenotype of an organism, but you want to know the genotype, you should do DNA analysis.

1- Breed the organism with another organism that you know the genotype to, and it should be homozygous recessive.

2- If all the offspring came out of the dominant allele's phenotype, then the genotype of the unknown organism is homozygous dominant.

3- If half the offspring were of the dominant allele's phenotype, and half were of the recessive allele's phenotype, then the genotype of unknown organism is heterozygous dominant.